Agentic AI

Estimated Reading Time: 15 minutes

Introduction: The Shifting Landscape of AI

The release of ChatGPT in late 2022 marked a turning point in how we think about artificial intelligence. At first, large language models (LLMs) could only generate text based on prompts. But when these models gained the ability to use tools, call functions, and access memory, they started to act more like agents.

AI agents are like digital assistants designed to carry out specific tasks—such as answering questions, organizing data, or summarizing content. But as demands grew more complex, a new approach emerged: Agentic AI.

Agentic AI refers to systems made up of multiple AI agents that work together. Instead of just reacting to one command, they can:

- Break down big goals into smaller tasks.
- Adapt to new inputs or situations.
- Communicate and coordinate with one another.

In short, Agentic AI moves beyond one-task agents and enables teams of AI working collaboratively to achieve more complex goals.

Defining the Spectrum: From AI Agents to Agentic AI

What are AI Agents?

AI Agents can be characterized as autonomous software entities designed for goal-directed task execution within specific digital environments. Their operation typically involves three capabilities:

- **Autonomy**: The ability to function with minimal human intervention after initial deployment, capable of perceiving environmental inputs, reasoning over contextual data, and executing actions in real-time.
- **Task-Specificity**: Each agent is optimized for narrow, well-defined tasks such as email filtering, database querying, and so on.
- **Reactivity**: Agents respond to inputs from users, APIs, or other software environments in real time.

This design enables applications such as customer service bots, internal document search assistants, and email triage tools.

The diagram below shows an AI agent that finds news, summarizes it, and sends it back.

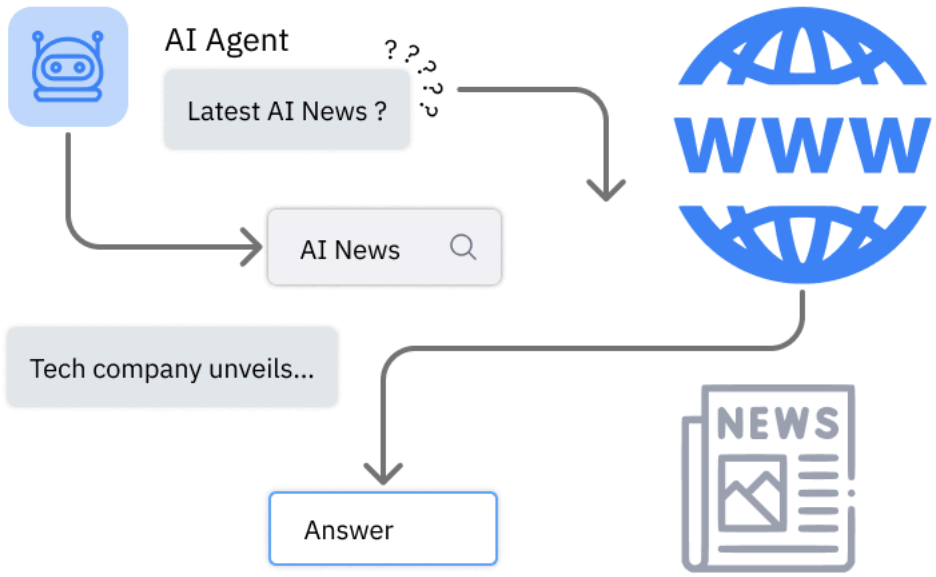

Image Reference: [AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges]

What Is Agentic AI?

Agentic AI takes things a step further. Instead of just one agent doing one job, Agentic AI brings multiple agents together into a team. These agents coordinate tasks, exchange information, adapt roles dynamically, and share memory. Key features include:

- **Task Decomposition**: Goals are split into subtasks automatically.
- **Inter-Agent Communication**: Agents share updates and results via messaging or shared memory.
- **Memory and Reflection**: Agents remember past steps and learn from outcomes.
- **Orchestration**: A lead agent or system coordinates the team.

Example: Planning a vacation—one agent books the flight, another finds hotels and a third checks visa requirements. A coordinator agent makes sure everything matches your preferences.

The following illustration contrasts a standalone AI Agent with a collaborative Agentic AI system in a smart home scenario.

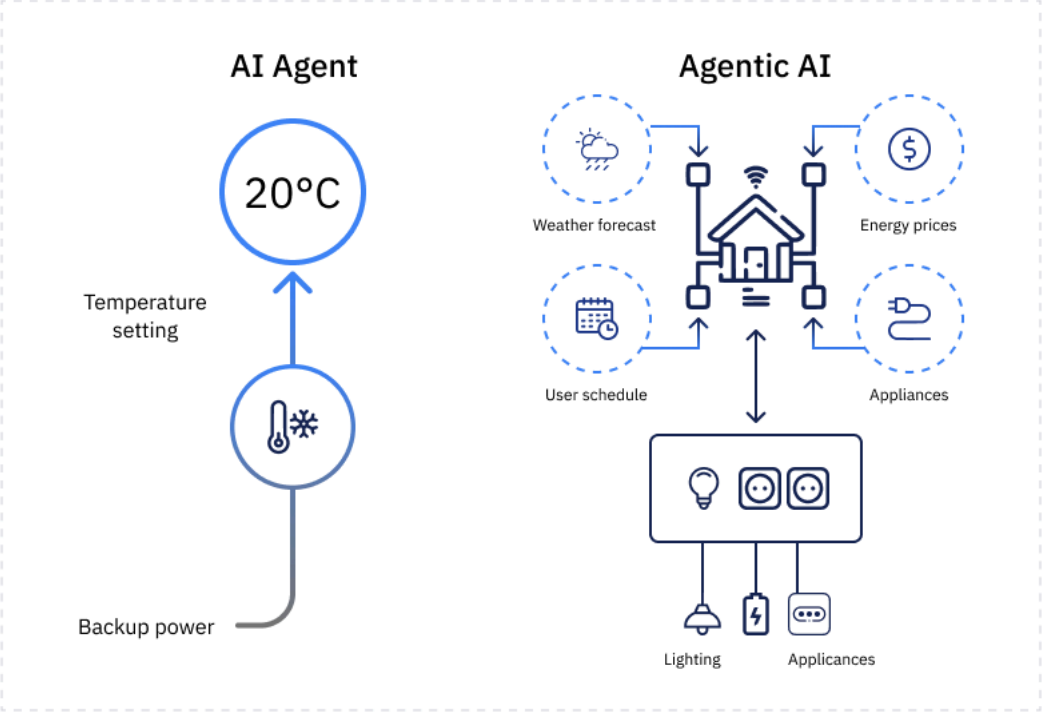

Image Reference: [AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges]

Key Architectural Differences between an AI Agent and Agentic AI

A structured taxonomy helps clarify the differences:

- **AI Agent**: Single entity, handles one task, uses tools, operates in narrow contexts
- **Agentic AI**: Multi-agent systems, goal-driven orchestration, adapts across time and context, supports parallel task execution

Feature	AI Agent	Agentic AI
Design	One agent, one task	Multiple agents with distinct roles
Communication	No coordination with others	Constant communication and coordination
Memory	Stateless or minimal history	Persistent memory of tasks, outcomes, and strategies
Reasoning	Linear logic (do step A → B)	Iterative planning and re-planning with advanced reasoning
Scalability	Limited to task size	Can scale to handle multi-agent, multi-stage problems
Typical Applications	Chatbots, virtual assistants, workflow helpers	Supply chain coordination, enterprise optimization, virtual team leaders

The architectural diagram below illustrates the transition from traditional AI Agent design (perception → reasoning → action) to Agentic AI systems built around collaboration, planning, and memory.

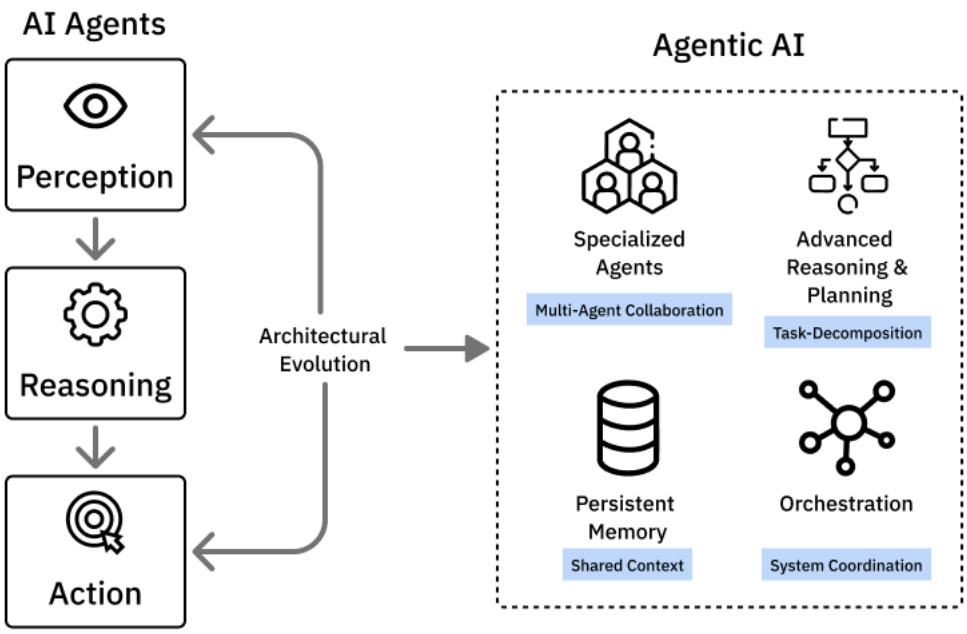

Image Reference: [AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges]

The transition from AI Agents to Agentic AI involves several fundamental architectural enhancements:

From Single to Multiple Agents:

- Rather than operating as single units, Agentic AI systems consist of multiple agents, each assigned specialized functions or tasks (such as summarization, retrieval, or planning).
- These agents interact via communication channels like message queues, blackboards, or shared memory.

Advanced Reasoning Capabilities

- Agentic systems integrate iterative reasoning capabilities using frameworks such as "ReAct (Reasoning and Acting)", "Chain-of-Thought" prompting, and "Tree of Thoughts." These mechanisms allow agents to break down complex tasks into multiple reasoning stages, evaluate intermediate results, and re-plan actions dynamically.

Persistent Memory Systems

- Unlike traditional agents, Agentic AI incorporates memory subsystems to preserve and persist knowledge across task cycles or agent sessions.
- Memory types include episodic memory (task-specific history), semantic memory (long-term facts or structured data), and vector-based memory for retrieval-augmented generation.

Real-World Applications

AI Agents excel in domains such as:

- Customer support (e.g., chatbots)
- Internal enterprise search
- Email filtering and prioritization

Agentic AI enables:

- Multi-agent research assistants
- Robotic coordination (e.g., drones)
- Collaborative medical decision systems
- Adaptive workflow automation

Current Challenges

Limitations of AI Agents

- AI Agents face significant challenges, including lack of causal understanding, inherited limitations from LLMs such as hallucinations and prompt sensitivity, incomplete agentic properties, and failures in long-horizon planning and recovery.

Agentic AI Complexities

- Agentic AI systems introduce amplified challenges, including inter-agent error cascades, coordination breakdowns, emergent instability, scalability limits, and explainability issues stemming from the complexity of orchestrating multiple agents across distributed tasks.

The Path Forward: Emerging Solutions

The field is actively developing solutions to address these limitations:

Retrieval-Augmented Generation (RAG)

- For AI Agents, RAG mitigates hallucinations and expands static LLM knowledge by grounding outputs in real-time data. In Agentic AI systems, RAG serves as a shared grounding mechanism across agents, allowing distributed agents to operate on a unified semantic layer.

Tool-Augmented Reasoning

- AI Agents benefit from function/tool calling, which extends their ability to interact with real-world systems. For Agentic AI, function calling is instrumental in enhancing both autonomy and structured coordination among multiple agents through orchestrated pipelines.

Memory Architectures

- Advanced memory systems address limitations in long-horizon planning by persisting information across tasks. Episodic memory allows agents to recall prior actions and feedback, semantic memory encodes structured domain knowledge, and vector memory enables similarity-based retrieval.

Looking Ahead: The Future of AI Agents and Agentic AI

AI Agents Evolution

AI agents are becoming smarter, so they will soon:

- Proactively recommend actions (not just react).
- Learn from interactions over time.
- Reason about more complex logic and decisions.

Agentic AI Advancement

It is expected that in the near future there will be:

- Teams of agents managing real-world workflows.
- Better simulations for testing multi-agent plans.
- Governance systems to ensure ethical coordination.

The diagram below outlines the anticipated evolution of AI Agents and Agentic AI systems across multiple dimensions.

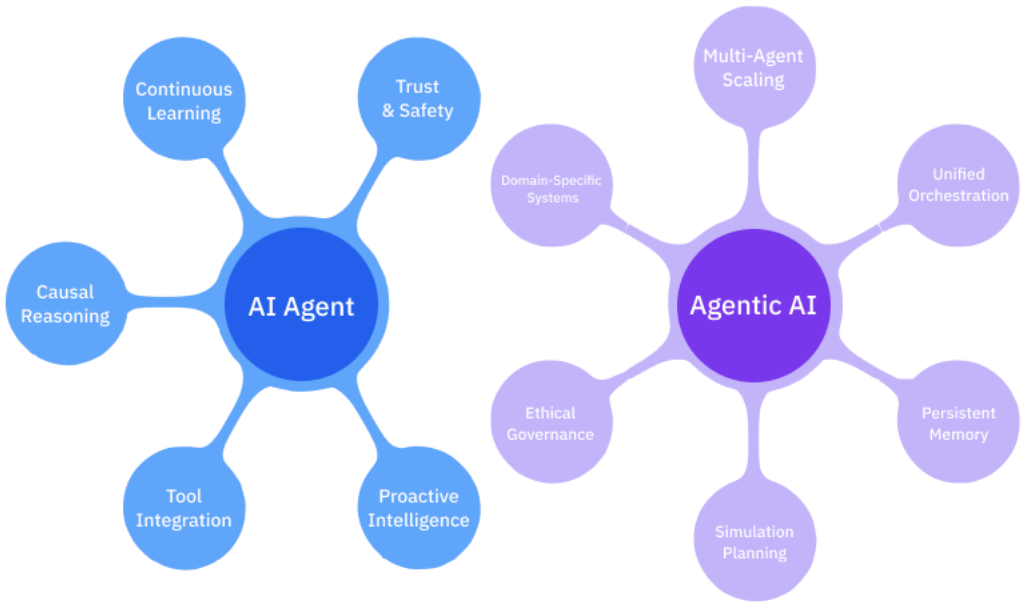

Image Reference: [AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges]

Building Agentic AI in Practice: Tools and Frameworks

While the concept of Agentic AI is still evolving, developers and researchers are already prototyping these systems using the following emerging AI orchestration frameworks:

- **LangChain**
  A Python framework for building applications around LLMs. It supports tool usage, memory, chains of reasoning, and agent interfaces. LangChain provides the building blocks to combine language models with external data and APIs.

- **LangGraph**
  A framework for building multi-agent workflows using a graph-based execution model. It allows you to define agents as nodes and their interactions as edges, ideal for orchestrating collaborative agents in Agentic AI.

- **IBM Bee, CrewAI, AutoGen, and others**
  These open-source tools simplify the design of multi-agent teams, role assignment, and structured task planning. They allow developers to simulate or deploy collaborative agent environments using memory, messaging, and dynamic delegation.

These frameworks allow developers to begin experimenting with Agentic AI by leveraging reusable modules for planning, reflection, and inter-agent communication.

Conclusion: An Evolving Paradigm

The distinction between AI Agents and Agentic AI is not static. These definitions continue to evolve as architectures, tools, and expectations shift. The future likely involves hybrid frameworks—blending the simplicity of task-specific agents with the flexibility and intelligence of multi-agent orchestration.

Agentic AI represents the next step in scalable, intelligent systems capable of real-world impact across domains such as robotics, healthcare, logistics, and beyond.

This reading material has taken inspiration from: AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges, which provides a comprehensive analysis of the evolving landscape of agent-based artificial intelligence systems.

Author(s)

Karan Goswami

Other Contributor(s)

Brian Ho
Faranak Heidari